
Machine Learning A

2026-2027

Home Assignment 2

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The deadline for this assignment is **17 September 2026, 22:00**. You must submit your *individual* solution electronically via the Absalon home page.

A solution consists of:

- A PDF file with detailed answers to the questions, which may include graphs and tables if needed. Do *not* include your full source code in the PDF file, only selected lines if you are asked to do so.
- A .zip file with all your solution source code with comments about the major steps involved in each question (see below). Source code must be submitted in the original file format, not as PDF. The programming language of the course is Python.
- **IMPORTANT: Do NOT zip the PDF file**, since zipped files cannot be opened in *SpeedGrader*. Zipped PDF submissions will not be graded.
- Your PDF report should be self-sufficient. I.e., it should be possible to grade it without opening the .zip file. We do not guarantee opening the .zip file when grading.
- Your code should be structured such that there is one main file (or one main file per question) that we can run to reproduce all the results presented in your report. This main file can, if you like, call other files with functions, classes, etc.
- Handwritten solutions will not be accepted. Please use the provided LaTeX template to write your report.
- You must adhere to the notations used in the task and in the lectures. If you need to use additional notations, ensure they are clearly and precisely defined. Undefined notations will result in point deduction.

1 Preprocessing (35 points) [Christian, W2 Monday]

To solve this exercise, it is helpful to read Section 9.1 in e-Chapter 9 of the book by Abu-Mostafa et al. (2015). It is also recommended to read Section 4.2 of the textbook (Abu-Mostafa et al., 2015) on regularization. The e-Chapter 9 can be downloaded from <https://amlbook.com/eChapters.html> or via Absalon. Note that the *in-sample error* E_{in} corresponds to what we call the empirical risk (or training error).

1.1 Importance of Preprocessing (8 points)

Each client at the Bank of Learning (BoL) is described by the two-dimensional feature vector (Age, Income). The bank gave two persons A and B credit cards based on their age and income.

Person	Age in years	Income in thousands of USD
A	47	35
B	22	40

A paid off her credit card bill, but B defaulted. Now consider C who has ‘coordinates’ (21, 36) applies for credit. Should the BoL give him credit, according to the nearest neighbor algorithm? If income is measured in ‘dollars’ instead of in ‘thousands of dollars’, what is your answer?

1.2 Input Centering [9 points]

Let $\mathbf{X} \in \mathbb{R}^{N \times d}$ be the data matrix, whose rows are formed the input data vectors $\mathbf{x}_1, \dots, \mathbf{x}_N$, all living in $\mathbb{R}^d$ (not augmented with a 1):

$$\mathbf{X} = \begin{bmatrix} \text{---} \mathbf{x}_1^\top \text{---} \\ \text{---} \mathbf{x}_2^\top \text{---} \\ \vdots \\ \text{---} \mathbf{x}_N^\top \text{---} \end{bmatrix}$$

Let $\bar{\mathbf{x}}$ be the *in-sample mean vector* of the input data, defined as $\bar{\mathbf{x}} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n$; in matrix notation, $\bar{\mathbf{x}} = \frac{1}{N} \mathbf{X}^\top \mathbf{1}$, where $\mathbf{1}$ is a vector of all ones. To obtain the transformed vector, we simply subtract the mean from each data point: $\mathbf{z}_n = \mathbf{x}_n - \bar{\mathbf{x}}$ for $n = 1, \dots, N$. Let $\mathbf{Z}$ be the matrix collecting $\mathbf{z}_1, \dots, \mathbf{z}_N$, constructed similarly to $\mathbf{X}$.

- (a) Define $\gamma = \mathbf{I} - \frac{1}{N}\mathbf{1}\mathbf{1}^\top$, where $\mathbf{I}$ denotes the identity matrix. Show that $\mathbf{Z} = \gamma\mathbf{X}$.

It is worth noting that γ is called the centering operator, which projects onto the space orthogonal to $\mathbf{1}$. You may read further in Appendix B.3 of Abu-Mostafa et al. (2015).

- (b) Show that the rank of $\mathbf{Z}$ is strictly smaller than N . You may presume the fundamental result that the column rank and the row rank are always equal.

1.3 Input Whitening [18 points]

Let $\hat{x}_1$ and $\hat{x}_2$ be independent with zero mean and unit variance. You measure inputs $x_1 = \hat{x}_1$ and $x_2 = \sqrt{1 - \epsilon^2}\hat{x}_1 + \epsilon\hat{x}_2$ for $\epsilon \in [-1, 1]$. Let $\mathbf{x} = (x_1, x_2)^\top$ and $\hat{\mathbf{x}} = (\hat{x}_1, \hat{x}_2)^\top$.

- (a) What are $\text{Var}(x_1)$, $\text{Var}(x_2)$, and $\text{Cov}(x_1, x_2)$? (Here, ‘Var’ and ‘Cov’ denotes the variance and covariance, respectively.)
- (b) Suppose $f(\hat{\mathbf{x}}) = \hat{w}_1\hat{x}_1 + \hat{w}_2\hat{x}_2$ (linear in the independent variables). Show that f is linear in the correlated inputs, $f(\mathbf{x}) = w_1x_1 + w_2x_2$. (Obtain w_1 , w_2 as functions of $\hat{w}_1$, $\hat{w}_2$.)
- (c) Consider the ‘simple’ target function $f(\hat{\mathbf{x}}) = \hat{x}_1 + \hat{x}_2$. If you perform regression with the correlated inputs $\mathbf{x}$ (i.e., your model is of the form $f(\mathbf{x}) = w_1x_1 + w_2x_2$) and regularization constraint $w_1^2 + w_2^2 \leq C$, what is the maximum amount of regularization you can use (i.e., the minimum value of C) and still be able to implement the target?
- (d) What happens to the minimum C as the correlation increases (i.e., $\epsilon \rightarrow 0$)?

Note: The question is adopted from Exercise 9.4 on page e-Chap:9–4 in the textbook by Abu-Mostafa et al. (2015). As in the book, the notation is overloaded. Part (b) could be rephrased as: Let $\mathbf{x} = (x_1, x_2)^\top$ depend on $\hat{\mathbf{x}} = (\hat{x}_1, \hat{x}_2)^\top$ as defined above. Let $f(\hat{\mathbf{x}}) = \hat{w}_1\hat{x}_1 + \hat{w}_2\hat{x}_2$ for some $w_1, w_2 \in \mathbb{R}$. Show that there exists a function $g(\mathbf{x}) = w_1x_1 + w_2x_2$ such that $f(\hat{\mathbf{x}}) = g(\mathbf{x})$.

Hints: Recall that for random variables X, Y, V, W and scalars a, b, c, d :

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y) \quad (1)$$

$$\begin{aligned} \text{Cov}(aX + bY, cV + dW) &= ac \text{Cov}(X, V) + ad \text{Cov}(X, W) \\ &\quad + bc \text{Cov}(Y, V) + bd \text{Cov}(Y, W) \end{aligned} \quad (2)$$

2 Hoeffding's Bound (15 points) [Sadegh, W3 Monday]

Consider a random variable X that takes values in $\{-2, 0.8, 1\}$. Assume all of the values are equiprobable, that is, $\mathbb{P}(X = -2) = \mathbb{P}(X = 0.8) = \mathbb{P}(X = 1)$.

We generate 100 samples of X in an i.i.d. fashion, and let S denote their sum. Use Hoeffding's inequality to bound the probability $S > 22$.

3 Illustration of Markov's, Chebyshev's, and Hoeffding's Inequalities (25 points) [Sadegh, W3 Monday]

3.a Make 1,000,000 repetitions of the experiment of drawing 20 i.i.d. Bernoulli random variables $X_1, \dots, X_{20}$ with mean 0.35 and answer the following questions.

1. Plot the empirical frequency of observing $\frac{1}{20} \sum_{i=1}^{20} X_i \geq \alpha$ for $\alpha \in \{0.5, 0.55, 0.6, \dots, 0.95, 1\}$.
2. Explain why the above granularity of α is sufficient. I.e., why, for example, taking $\alpha = 0.51$ will not provide any extra information about the experiment.
3. In the same figure plot the Markov's bound¹ on $\mathbb{P}(\frac{1}{20} \sum_{i=1}^{20} X_i \geq \alpha)$.
4. In the same figure plot the Chebyshev's bound² on $\mathbb{P}(\frac{1}{20} \sum_{i=1}^{20} X_i \geq \alpha)$. (You may have a problem calculating the bound for some values of α . In that case and whenever the bound exceeds 1, replace it with the trivial bound of 1, because we know that probabilities are always bounded by 1.)
5. In the same figure plot the Hoeffding's bound³ on $\mathbb{P}(\frac{1}{20} \sum_{i=1}^{20} X_i \geq \alpha)$.
6. Compare the four plots.
7. For $\alpha = 1$ and $\alpha = 0.95$ calculate the exact probability $\mathbb{P}(\frac{1}{20} \sum_{i=1}^{20} X_i \geq \alpha)$. (No need to add this one to the plot.)

3.b [Optional, 0 points] Repeat the question with $X_1, \dots, X_{20}$ with mean 0.1 (i.e., $\mathbb{E}[X_1] = 0.1$) and $\alpha \in \{0.1, 0.15, \dots, 1\}$.

Do not forget to put axis labels and a legend in your plot.

¹Markov's bound is the right hand side of Markov's inequality.

²Chebyshev's bound is the right hand side of Chebyshev's inequality.

³Hoeffding's bound is the right hand side of Hoeffding's inequality.

4 The Role of Independence (optional, 0 points)

[Sadegh, W3 Monday]

Design an example of identically distributed, but *dependent* Bernoulli random variables $X_1, \dots, X_n$ (i.e., $X_i \in \{0, 1\}$), such that

$$\mathbb{P}\left(\left|\mu - \frac{1}{n} \sum_{i=1}^n X_i\right| \geq \frac{1}{2}\right) = 1,$$

where $\mu = \mathbb{E}[X_1]$.

Note that in this case $\frac{1}{n} \sum_{i=1}^n X_i$ does not converge to μ as n goes to infinity. The example shows that independence is crucial for convergence of empirical means to the expected values.

References

Yaser S. Abu-Mostafa, Malik Magdon-Ismail, and Hsuan-Tien Lin. *Learning from Data. Dynamic E-Chapters*. AMLbook, 2015.